A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 07

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1 Gravimetry:** variations in gravity caused by differences in the density and mass of the rock below – ore-bearing rocks are usually denser than the surrounding sedimentary deposits **1 Magnetometry:** rocks that are more magnetic than their surroundings, such as the iron ore magnetite and the ores of tungsten and cobalt **1 Seismic surveys:** echoes of sound waves from controlled explosions or a surface vibrator, giving the depth, density and shape of the rock strata **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Trial drilling is the most expensive technique per sampling site, so the remote surveys are used first to narrow the search to the most promising ground **1** Drilling is the only technique that brings up actual samples of the rock, so it is used to confirm the composition and purity found by chemical analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reserves are the deposits that can be extracted economically with current technology – the resource base is everything that exists, including deposits that are too low-grade, too deep or too remote to pay **1** A higher price lowers the cut-off ore grade, so deposits that previously could not be mined at a profit become worth extracting and are reclassified as reserves **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** All the rock above the ore (the overburden) must be stripped off, so habitat is destroyed over a much larger area than with deep mining **1** Spoil heaps of waste rock occupy land, may be unstable and can be colonised only slowly because they hold few nutrients and little water **1** Dust from blasting, crushing and haulage raises atmospheric particulate levels and settles on surrounding vegetation **1** Noise and vibration from blasting and heavy machinery disturbs wildlife and nearby residents **1** Drainage from the workings and spoil is often acidic and carries dissolved metals, polluting watercourses and groundwater **1** Lowering the water table to keep the pit dry dries out nearby wetlands and wells **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4

Do not write outside the box

Question	Marking guidance	Marks
01.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Topsoil and subsoil stripped at the start are stored separately and replaced in the correct order, so the restored soil has a fertile surface layer **1** The land is re-contoured and spoil heaps are given shallow, stable slopes before being covered with soil and planted **1** The site can be returned to agriculture or forestry, or flooded workings developed as a wetland or water-sport habitat **1** Planning restoration first means the topsoil is saved rather than buried, the final landform is built as mining proceeds, and the cost is included in the economic assessment of the mine **1** A restoration bond or condition in the planning permission makes certain the work is paid for even if the company fails **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 136-142; AQA spec 3.2.3.4.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Most energy comes from non-renewable fossil fuels whose reserves are depleting – the rate of new discoveries has fallen sharply since the 1980s **1 Use causes unacceptable environmental damage:** atmospheric pollution, climate change, oil pollution, habitat loss **1 Economic / social:** depleting resources become more expensive, reducing future affluence – risk of an energy gap if new technologies are not developed in time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3
02.2	**1 mark each** 1 mark for each of the 4 points below. **1 Coal mining:** habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise **1 Oil and gas extraction:** spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO2 / CO2 / smoke **1** Pipelines causing habitat damage during construction – tanker transport risks **1** Fuel processing (refineries, coal washing) pollution – embodied energy of equipment and site development for all resources **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	4

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) **1** Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise **1** By then depletion would reduce supply faster than replacements could be built, causing shortages **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Surplus energy drives a pump compressing air stored in underground caverns (salt mines) – releasing it drives machinery / turbines **1** Compression produces a lot of heat – storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** As volume increases the surface area : volume ratio decreases **1** Heat is lost through the surface so there is less surface per unit of stored heat – very large stores can carry summer solar heat into winter (inter-seasonal storage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Potassium nitrate has a much higher boiling point so it stores heat at up to 550 °C without pressurisation **1** Water could only reach such temperatures under high pressure, increasing costs – high temperature needed to raise steam for turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Materials with high specific heat capacity (concrete, water) warm and cool slowly **1** They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Flywheels:** rotating mass stores kinetic energy for short-term smoothing / generation **1 Supercapacitors:** electrochemical capacitors with 10-100 times the energy density of conventional capacitors – possible future large-scale electricity storage **1** Thermal storage in molten salt / high-volume tanks – hydrogen / P2G – batteries **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **Critical Group** → the people most at risk / most heavily exposed	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate **1** No need to sample sites that were never likely to be polluted – sampling also checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify **1** So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
05.1	**1 mark each** 1 mark for each of the 6 points below. **1** Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes **1** Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH **1** Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow **1** Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser **1** Unattractive topography - landscaping / reshaping **1** Dust / heavy metals - cover and keep at high pH **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leachate** → polluted liquid draining out of waste **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	6
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** So that plant roots cannot reach the toxic layer **1** Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions **1** Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	
06.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1

Do not write outside the box

Question	Marking guidance	Marks
06.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Number all possible sampling sites (eg on a grid map) **1** Select the sites using a random number generator / table – (regular shape: grid coordinates from random numbers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis **1** So the sample does not represent the whole area and conclusions are invalid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** Screens (metal grills) trap floating / suspended non-biodegradable items (paper, plastic) which are landfilled or incinerated **1 Grit traps:** the channel widens so flow slows and stones / road grit sink out because there is not enough kinetic energy to carry them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers	2
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** The effluent stands in large tanks so the faecal / organic solids sink and are removed as sludge **1** This removes over 95% of the organic matter from the fluid so the risk of deoxygenation of the river is greatly reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deoxygenation** → loss of dissolved oxygen from the water	2
08.3	**1 mark each** 1 mark for each of the 3 points below. **1** Aerobic bacteria digest / break down the remaining dissolved and suspended organic matter **1** Air (or pure oxygen) is mixed in by paddle wheels or air stones so oxygen shortage does not limit the aerobic process **1** The suspended bacteria collected in the secondary sedimentation tank are the 'activated sludge' – returning them maintains a large bacterial population so digestion is rapid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.4	**1 mark each** 1 mark for each of the 3 points below. **1** Effluent from secondary treatment may still contain dissolved inorganic nutrients (phosphates, nitrates) that cause eutrophication, and some bacteria **1** Needed where the discharge site is ecologically sensitive or used by many people (bathing water, drinking water abstraction) **1** Phosphate stripping with iron(III) sulfate – micro-strainers (very fine sieves) then UV light or chlorine to kill remaining bacteria **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth • **abstraction** → water taken out of a river, lake or aquifer **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 269-272; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 Overgrazing:** vegetation is eaten faster than it grows so soil is exposed – hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff **2** Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles – fewer worms so less aeration and lower infiltration capacity – fewer nutrients released so less vegetation cover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **humus** → dark, decomposed organic matter in soil **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Tree roots that held the soil together decompose so the soil is bound less strongly **1** Heavy rain adds mass and lubricates particles – without interception more water enters the soil, making slumping more likely **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Sedimentation in rivers reduces flow capacity so flooding downstream is more likely **1** Sedimentation in reservoirs reduces their storage volume **1** Coastal sedimentation covering reefs / seagrass – turbidity reducing light for aquatic plants **1** Increased atmospheric particulates from wind erosion worsening asthma – dust covering crops / urban areas **1** Desertification as vegetation cannot re-establish – reduced rainfall **1** Nutrients (phosphates) carried on soil particles causing eutrophication **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
10.1	**1 mark each** 1 mark for each of the 3 points below. **1** Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) **1** They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone **1** The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3

Do not write outside the box

Question	Marking guidance	Marks
10.2	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Provides carbon, oxygen, hydrogen and nitrogen (as N_2, O_2, CO_2, H_2O) for biological molecules **1** Delays the escape of infrared energy, keeping the surface warm **1** Distributes heat and water vapour by winds / ocean currents **1** Provides atmospheric pressure so liquid water can exist **1** Deflects damaging radiation of the solar wind in the upper atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
10.3	1 mark for the point below. **1** Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **stratified** → sampling each area or group in proportion to its size	1

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Troposphere:** enhanced greenhouse effect - anthropogenic CO_2 (fossil fuels, deforestation), methane (rice, livestock, landfill), N_2O, CFCs, tropospheric ozone – more infrared absorbed and re-emitted – aerosols and smoke increase albedo / scatter light – land-use change alters surface albedo and evapotranspiration • **Stratosphere:** CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium – less UV absorbed in the stratosphere, more reaches the surface – stratospheric cooling • **Consequences:** changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions – increased UV damage to organisms • **Difficulties of prediction:** linked processes where one change alters others – feedback (positive and negative) and tipping points – time lags (ocean heat capacity, long residence times of CFCs and CO_2) – natural variability (solar output, volcanoes, Milankovitch cycles) – incomplete understanding of atmospheric processes – limited historical data so proxies needed – models with different assumptions • **Evaluation:** some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain – precautionary principle • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **enhanced greenhouse effect** → extra warming caused by added greenhouse gases • **anthropogenic** → human-caused / caused by human activity • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	In the troposphere human activity enhances the greenhouse effect: fossil fuels and deforestation add CO_2, rice, livestock and landfill add methane, fertilisers add nitrous oxide, and CFCs and tropospheric ozone are new absorbers. More outgoing infrared is absorbed and re-emitted, warming the surface, while smoke and aerosols scatter sunlight and land-use change alters albedo and evapotranspiration. In the stratosphere, CFCs, halons and HCFCs release chlorine and bromine that catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium so that less UV is absorbed there and more reaches the surface, cooling the stratosphere as well.	
	The consequences run through every system the atmosphere's energy drives: climate, thermohaline circulation, the hydrological cycle, sea level and species distributions, plus UV damage to organisms. Predicting them is hard because the processes are linked - one change alters others - and because feedback can amplify (albedo loss, methane release) or damp (cloud cover, photosynthesis) the effect, sometimes past tipping points. Time lags from ocean heat capacity and the long residence times of CO_2 and CFCs, natural variability from solar output, volcanoes and orbital cycles, sparse historical data replaced by proxies, and models built on different assumptions all add uncertainty.	
	Some things are nevertheless well established: the physics of infrared absorption, the chemistry of ozone loss, and the success of the Montreal Protocol. What is uncertain is the scale and timing of regional impacts, which is exactly why the precautionary principle, not certainty, should guide action.	
	Weaker answer - Level 1 (3 marks)	
	Humans release greenhouse gases like carbon dioxide from burning fossil fuels into the troposphere, which traps more heat and causes global warming. CFCs go into the stratosphere and destroy the ozone layer, so more UV reaches the Earth. It is hard to predict what will happen because the atmosphere is complicated and scientists do not know everything about it. There could be feedback effects that make it worse.	
	What makes the difference	
	• Level 1 gives one cause per layer; Level 3 lists several gases and processes and explains the mechanism (absorption and re-emission; catalytic destruction shifting an equilibrium).	
	• 'Difficulties of prediction' is answered in Level 3 with specific reasons (linked processes, feedback, time lags, natural variability, proxy data, models); Level 1 says 'complicated'.	
	• Level 3 evaluates - separating what is certain from what is not - and links this to the precautionary principle.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Problematic properties of solar:** intermittency, unreliability from cloud, low energy density, changing angle – locational constraints (deserts, latitude) • **Harnessing solar:** photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms – environmental impacts of manufacture and land use • **New solar technologies:** multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • **HEP:** gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links – combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	

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